Siddharth Patel

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Research Interests: Hardware for AI & Computer architecture · Digital VLSI & SoC design · Memory systems & compute-in-memory · Efficient & adaptive AI inference and training

Education

Shiv Nadar Institution of Eminence — Greater Noida, India
B.Tech, Electrical & Computer Engineering — *Aug 2023 – Expected May 2027*

- **CGPA: 8.75/10.0 · Rank: Top 10% of cohort**
- **Student of the Year, Class of 2027 – university-wide recipient**
- **Dean's List – Fall 2023, Spring 2024, Fall 2025 & Spring 2026**
- *Specialization:* Nanoelectronics & Embedded Systems
- *Undergraduate Thesis:* "An Open-Source GPU Microarchitecture: A Patent-Unencumbered Design with Synthesizable RTL" – in progress (2026–2027)
- *Selected coursework:* ASIC: RTL-to-GDS, VLSI Design, Digital Design with FPGAs, Computer Organization & Design, Semiconductor Devices, Deep Learning, Large Language Models, Embedded Systems

Publications

*(author in **bold**; † corresponding author)*

Journal Articles

1. **S. Patel**† and R. Singh, "AI Hardware Accelerators for Large Language Models: Architectures and the Memory Wall," *ACM Computing Surveys*, 2026. **[Under review]** *arXiv preprint*, 2026. arXiv:2608.28048

2. **S. Patel**†, A. Awasthi, A. Chaurasia, and R. Singh, "Bitwise-Reproducible Reduction Across Cluster Reshape: A Compiler/Fabric ABI," *IEEE Transactions on Parallel and Distributed Systems*, 2026. **[Under review]**

3. **S. Patel** and R. Singh, "RMAI: Runtime Multi-Objective Adaptive Inference through Joint Dynamic Sparsity and Precision Scaling on Resource-Constrained IoT Edge Devices," *IEEE Internet of Things Journal*, 2026. **[Under review]**

Conference Papers

4. **S. Patel**† and R. Singh, "Hardware-Enabled Root of Trust for IoT Security," in *Proc. IEEE Intl. Conf. Intelligent Systems and Embedded Design (ISED)*, 2025. **[Published]** DOI: 10.1109/ISED67359.2025.11405039

5. S. V. Rajeshwaran, O. Singh, **S. Patel**, S. Singhal, and R. Singh, "Design and Implementation of a Smart Power Controller for Sensor Nodes," in *Proc. Intl. Conf. on Intelligent Technologies for a Sustainable & Inclusive Future (ICITSIF)*, 2026. **[Published]**DOI: 10.1109/ICITSIF69060.2026.11608698

6. **S. Patel**†,V. Singh, S. Malu and R. Singh, "Learn the Interface, Not the Implementation: Output Representation Dominates Training Regime in LLM-Based RTL Generation," Submitted to *40th IEEE International Conference on VLSI Design (VLSID 2027)*. **[Under Review]**

Research Experience

An Open-Source GPU Microarchitecture (Undergraduate Thesis) — 2026 – Present
Undergraduate Researcher · Advisor: Dr. Rohit Singh — *Greater Noida, India*

- Designing a fully open, patent-unencumbered GPU from first principles – microarchitecture, ISA, and synthesizable RTL – to address the absence of an open, industrial-grade GPU (existing designs are locked behind AMD/NVIDIA IP).
- Releasing as open source for research and educational use.

Hardware for AI — Deterministic Large-Model Inference Dec 2025 – Present
Undergraduate Researcher · Advisors: Dr. Rakesh Palishtty & Dr. Rohit Singh *Greater Noida, India*

- Investigating sources of non-determinism in large-model inference on parallel hardware, and co-designing compiler- and fabric-level mechanisms to eliminate them.
- Designed a compiler/fabric ABI that **guarantees bitwise-reproducible reductions across heterogeneous GPU cluster reshapes** – bit-identical results independent of how the cluster is partitioned or resized. *(→ IEEE TPDS [under review])*
- Prototyping compute-in-memory primitives and alternative number systems for deterministic computation.

Edge-AI Optimization — Adaptive & Architecture-Aware Inference Oct 2025 – May 2026
Undergraduate Researcher · Advisor: Dr. Rohit Singh *Greater Noida, India*

- Designed **RMAI**, a runtime adaptive-inference framework that jointly tunes per-layer weight sparsity and quantization precision to live device state (battery, thermal, memory). *(→ IEEE IoT Journal [under review])*
- Demonstrated up to **14.8% lower latency and 15.4% energy savings** on ResNet-50 vs. a static INT8 baseline (Raspberry Pi 4), with **38× greater thermal robustness** under sustained load; validated across ResNet-50, MobileNetV2, DeiT-Small, and YOLOv8m.
- Developed **architecture-aware pruning and quantization for CPU-only edge inference**.

Smart Power Controller for Sensor Nodes Dec 2024 – Oct 2025
Research Assistant · Advisor: Dr. Rohit Singh *Greater Noida, India*

- Contributed to the design of an energy-efficient wireless sensor node with a programmable power-control unit for smart-irrigation / smart-city deployments.
- Implemented demand-based power management that **doubled battery life** for power-hungry sensor nodes. *(→ ICITSIF [published])*

Hardware-Enabled Root of Trust for IoT Aug 2023 – Dec 2024
Undergraduate Researcher · Advisor: Dr. Rohit Singh *Greater Noida, India*

- Designed a hardware root-of-trust with immutable device identities and RSA-OAEP secure boot to resist spoofing and tampering.
- Achieved **99.8% authentication accuracy** under simulated attack scenarios. *(→ IEEE ISED [published])*

Industry Experience

Technology R&D (TRD) Intern Jun 2026 – Aug 2026
STMicroelectronics *Greater Noida, India*

- Building **AI-driven automation for STMicroelectronics' end-to-end memory-design workflow**, replacing a manual, multi-engineer, per-variant RTL process with an intelligent generation pipeline.
- Applying **ML-guided design-space exploration and learned optimization**, and **fine-tuning LLMs** for domain-specific accuracy on RTL tasks.
- Developing **LLM-assisted RTL generation** and architecting a **domain-specific AI agent from scratch** to drive the flow end-to-end.
- Produces synthesis-ready, **ECC-protected memory subsystems** (encoder/decoder RTL; SEC/SECDED) across 7+ configuration axes, byte- to gigabyte-scale.

Independent Consultant — Electronics Design Feb 2025 – Jul 2025
[Electronics Manufacturer] *Delhi NCR, India*

- **Sole engineer** on a full product redesign – re-architected 3-/4-phase voltage stabilizers end-to-end (circuit, enclosure, PCB, firmware) and delivered independently, ahead of schedule.
- Cut bill-of-materials and assembly complexity via modular reuse (**∼70% lower production cost**), with backward compatibility for legacy systems.

Teaching Experience

Teaching Assistant — Digital Electronics Aug 2026 – Present
Department of Electrical Engineering, Shiv Nadar Institution of Eminence *Greater Noida, India*

– Instructing hands-on lab sessions for **60 students**; independently designing **three FPGA lab experiments** to teach Verilog through progressively harder RTL design tasks.

Curriculum Designer & Lab Instructor — Electronics & IoT Jul 2025 – May 2026
Department of Electrical Engineering, Shiv Nadar Institution of Eminence *Greater Noida, India*

– Designed lab experiments, modules, learning objectives, and assessments for the first-year Electronics & IoT curriculum (**480+ students**).

– Instructed **120+ students** in hands-on electronics / IoT lab sessions.

– Led build-out of a 20-station Electronics Research & Innovation Lab (equipment procurement + setup).

Honors & Awards

Student of the Year, Class of 2027 – sole university-wide recipient across all departments, Shiv Nadar Institution of Eminence (2026)

Selected from 20,000+ applicants nationwide – Anveshana National Science Fair (2025)

Atal Incubation Centre Scholar – competitively selected for the *Life Link* health platform; awarded a grant (INR 100K) with mentorship and incubation support (2024–25)

Invited Peer Reviewer – IEEE Intl. Conf. on Next-Generation Adaptive Research & Innovations (ICNARI 2026)

Mathematics Olympiad – Gold Medal (national level, 2019)

Technical Skills

Languages: Python, C/C++, Verilog, SystemVerilog, VHDL, Amaranth HDL, Tcl
Computer Architecture & Digital Design: RISC-V, CPU Microarchitecture, ASIC/RTL Design, FPGA Design, Parameterized RTL Generation
Memory & VLSI: SRAM/DRAM Architecture, Memory IP Development, Compute-in-Memory (CIM), ECC & Coding Theory (SEC/SECDED, BCH, Hamming), Formal Verification
AI/ML: PyTorch, Custom ML Library Development, Transformer/LLM Architectures, LLM Fine-Tuning, LLM-Assisted RTL Generation, AI Agents, ML-Guided Design-Space Exploration, Mixed-Precision Quantization, Dynamic Sparsity
EDA Tools: Synopsys (RTL-to-GDS flow), Cadence Virtuoso & Xcelium, Xilinx Vivado, KiCad
Embedded: STM32, Arduino, Raspberry Pi, Embedded Linux

Outreach

Anveshana National Science Fair – led STEM workshops for 50+ students and presented a secure-IoT prototype to 300+ attendees, Pragati Maidan, Delhi (2024–25).